

Sumit Kumar Chaudhary

Email:chaudharysumit130@gmail.com | Phone: +91-9304082560 | LinkedIn: sumitchaudhary14

EDUCATION

Year	Degree/Certificate	Institute	CPI/Percent
2022-2024	M.Tech CSE	Indian Institute of Technology, Kanpur	7.71/10
2017-2021	B.Tech CSE	MMMUT, Gorakhpur	7.47/10
2016	CBSE (XII)	Chinmaya Vidyalaya , Bokaro	86.6 %
2014	ICSE(X)	De Nobili School C.T.P.S. , Bokaro	93.17%

PROFESSIONAL EXPERIENCE

- SDE-1 | C3iHub, IIT-KANPUR** Jul'24 - Present
 - Architected and deployed a **Generative AI assessment microservice** using Docker and FastAPI, supporting policy compliance analysis with a Retrieval-Augmented Generation (RAG) pipeline
 - Reduced manual assessment time by **80%** and achieved **>85%** accuracy in compliance detection by leveraging semantic search and text embeddings for analyzing unstructured documents
 - Designed and owned the end-to-end data workflow, **creating robust API contracts** for asynchronous document ingestion, real-time status tracking, and structured JSON output for the frontend.
- Research Scholar Intern | Center for Developing Intelligent Systems(CDIS) IIT-KANPUR** Jan'24 - Mar'24
 - Developed a retrieval-augmented generation (RAG) pipeline in Python to answer natural language questions over a custom knowledge base (e.g., research papers).
 - Utilized sentence-transformer models for creating dense vector embeddings and implemented a FAISS vector index for highly efficient semantic search and retrieval.

PROJECTS

- CSCMM Cybersecurity Assessment Tool**
 - Led a **2-person team** to design and implement a full-stack Cybersecurity Maturity assessment platform for a Govt. of India Project.
 - Built the platform using a **ReactJS frontend, NodeJS backend, and PostgreSQL database**, featuring user authentication and role-based access control.
 - Implemented complex business logic, including weighted scoring for domains/processes, and automated data extraction from SQLite to CSV for compliance tracking.
- DeceptiSense: Cyber Attack Pattern Recognition (M.Tech Thesis)**
 - Engineered a **scalable, end-to-end data pipeline** in Python to process and analyze real-time malware artifacts captured from Dockerized honeypots.
 - Developed a **container-based data extraction module** with Docker-layer file extraction to automate malware artifact collection, significantly reducing incident response time.
 - Integrated the system with the **MITRE ATT&CK framework** using clustering and formal feature mapping for automated classification of malicious activities and threat intelligence generation.
- End-to-End OCR Pipeline for CAPTCHA Solving**
 - Built a complete Optical Character Recognition (OCR) pipeline involving image preprocessing (noise reduction, binarization), character segmentation using contour detection, and classification.
 - Explored character recognition using **SVM** and **Logistic Regression** and achieved **98.5%** accuracy with a compact 174KB model and quick (4ms) prediction time

TECHNICAL SKILLS

- Programming:** Python, C++, Javascript,C
- Web & Backend:** FastAPI, Node.js, React.js, REST APIs, PostgreSQL, MongoDB, MySQL
- Cloud & DevOps:** Docker, Microservice Architecture, Git, CI/CD Concepts, AWS (S3, EC2)
- AI/ML Systems:** Generative AI, RAG, Machine Learning, PyTorch, Scikit-learn, OpenAI API , Numpy , Pandas